

		motion of the curve path for the rider to bend left.
12	C	Centripetal force is the net force which produces centripetal acceleration. $T - mg\cos\theta = \frac{mv^2}{r} = \frac{2KE}{r}$ $T = mg\cos\theta + \frac{2KE}{r}$ Since the object is rotating freely, due to conservation of energy, the lower it is the more KE it has. At the lowest point, both terms on the RHS is maximum hence T is maximum. At the highest point, both terms on the LHS is minimum hence T is minimum.

2020

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

13

A

π

2π

2π

$2\pi r$